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Effects of Early Light-Quality Pretreatment During Withering on the Taste and Aroma Quality of Autumn White Tea

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ABSTRACT

This study establishes early light-quality pretreatment as an effective strategy to enhance the flavor of autumn white tea. A multi-platform framework integrating sensory evaluation, electronic tongue (E-tongue), electronic nose (E-nose), high-performance liquid chromatography (HPLC), ultraviolet-visible (UV-Vis) spectrophotometry, and gas chromatography-ion mobility spectrometry (GC-IMS) was employed to correlate sensory perception with chemical composition. Blue-light pretreatment markedly improved sweetness, umami, and a more coordinated aroma profile characterized by sweet/caramel-like and roasted notes, while reducing bitterness, astringent aftertaste, and green-note-related characteristics. Compared with red- and yellow-light pretreatments, blue light showed lower instrumental bitterness, and it also reduced residual aftertaste-related responses and green-note-related volatile characteristics. These changes were associated with increased total free amino acids and soluble sugars, altered catechin composition, and reduced green-note volatiles. Yellow light preferentially enhanced floral- and fruity-related volatile profiles, whereas red light showed limited overall benefits. Multivariate and Mantel analyses supported significant associations between chemical profiles and sensor responses, including epigallocatechin gallate (EGCG)–astringency and epicatechin gallate (ECG)–aftertaste-B. Collectively, blue-light pretreatment may improve autumn white tea flavor through coordinated changes in taste-active compounds and volatile metabolites, providing a practical basis for precision control of white tea quality.

1 | Introduction

White tea is one of the six major tea categories in China and originated in Fujian Province, which currently accounts for more than 90% of the national production (Ma et al. 2022). Owing to its mellow and sweet taste and its reported anti-inflammatory, anticancer, antibacterial, and antioxidant activities (Dias et al. 2013), white tea is highly appreciated by consumers. However, tea leaves harvested in summer and autumn are often characterized by excessive bitterness and astringency because high temperature and intense solar radiation promote the accumulation of bitter and astringent compounds (Gong et al. 2020). Recent studies have further shown that summer light and heat stress increase catechin accumulation in later-season tea leaves, thereby aggravating bitterness and astringency, whereas postharvest processing can only partially alleviate these sensory defects (S. S. Dong et al.

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2024; Qi et al. 2024). The resulting mismatch between supply and market acceptance has become a major constraint on the utilization of summer–autumn tea resources and has caused substantial economic losses in China (Han et al. 2021).

Improving the processing utilization of summer–autumn tea is therefore important for both sustainable tea industry development and growers' income. Among the manufacturing steps, withering is a critical stage in white tea processing because it drives extensive biochemical transformations that largely determine the final taste and aroma quality. Conventional withering methods include natural withering, sunlight withering, and heated-air withering (Qi et al. 2024). To improve processing efficiency and product quality, a range of modified withering strategies has been explored, including low-temperature freezing (Muthumani and Senthil Kumar 2007), CO_2/ethylene-assisted withering (Collings et al. 2021), and combined treatments. Among these approaches, regulation of light conditions during withering has received increasing attention as an environmentally friendly strategy for flavor improvement.

Accumulating evidence indicates that light during withering affects the biosynthesis of flavonoids, amino acids, and volatile compounds and thereby alters tea sensory quality. Light-assisted withering increases flavonoid glycosides and antioxidant activity in white tea (Zou et al. 2022). In addition, different LED wavelengths significantly influence aroma formation in tea (J. H. He et al. 2026; Mu et al. 2021; S. Y. Tian et al. 2024). Blue- and red-enriched spectra have been shown to suppress grassy notes while enhancing floral and fruity attributes by modulating volatile and metabolic pathways during withering (Hua et al. 2024). In white tea, spectral tailoring under LED illumination can direct the accumulation of flavor-related metabolites and shape sensory attributes during withering (Fan et al. 2025). Moreover, both spectral composition and light intensity have been shown to influence volatile profiles and aroma characteristics during postharvest leaf spreading (Y. Y. He et al. 2024, 2025). Recent studies also suggest that the effects of light can interact with processing operations; for example, gentle shaking combined with yellow-light pretreatment further enhances floral aroma formation in fermented tea, and composite LED spectra increase aroma diversity during black tea withering, compared with monochromatic light treatments (Ai et al. 2025; Y. F. Li et al. 2025). Beyond visible light, postharvest UV-B irradiation has also been reported to improve flavor quality in white and green teas through the regulation of key volatile pathways (Tan et al. 2025).

Although previous studies have demonstrated the potential of light regulation to improve tea quality, the comparative effects of red, yellow, and blue light on the taste and aroma quality of autumn white tea have not been systematically elucidated. In particular, the chemical basis underlying light-induced changes in sensory quality remains poorly understood.

Therefore, this study investigated the effects of light quality during withering on the taste and aroma quality of autumn white tea. Fresh leaves with one bud and two leaves were withered under natural light (CK) and red (R), yellow (Y), and blue (B) LED illumination. Sensory evaluation, electronic tongue (E-tongue), high-performance liquid chromatography (HPLC), electronic nose (E-nose) and gas chromatography-ion mobility spectrometry (GC-IMS) were used to characterize taste and aroma attributes and their associated chemical components. By integrating sensory data with multivariate analysis and Mantel analysis, this study established a framework linking light quality, digital sensory responses and chemical profiles, thereby providing a theoretical basis and technical support for optimizing autumn white tea processing and improving the utilization of summer–autumn tea resources.

2 | Material and Methods

2.1 | Sample Collection and Processing

Fresh tea leaves consisting of one bud and two leaves were harvested in late September from Jiangning District, Nanjing, China. To ensure consistent withering conditions, a custom-built light-controlled chamber with dimensions of 90 × 45 × 35 cm was used. The indoor temperature and relative humidity during withering were maintained at 24°C ± 3°C and approximately 65%, respectively. Red, yellow, and blue LED light strips were used for the light-quality treatments, each with a power of 18 W and a luminous flux of 1440 lm. Natural-light withering was conducted under indoor diffuse natural light, with an illuminance of approximately 800 lx.

Based on previous studies showing that tea leaves are sensitive to light quality during the early withering stage and that short-term light exposure can regulate tea metabolites and flavor-related compounds (Ai et al. 2017; Deng et al. 2020; M. Y. Li et al. 2022; Y. Li et al. 2023; J. Lin et al. 2022; Ye et al. 2018), 3.5 h was selected as an exploratory early-stage LED illumination period in this study. The LED pretreatment was initiated immediately after spreading. During the 3.5 h pretreatment, the leaves remained stationary, and no turning operation was performed; the LED strips and chamber configuration remained fixed throughout the treatment. This spread-leaf arrangement was comparable to conventional indoor natural withering of white tea, in which fresh leaves are laid on bamboo plaques or sieves and allowed to wither over a prolonged period (Z. Wang et al. 2024; Wu et al. 2022). After the LED pretreatment, all treated leaves were transferred to the same indoor diffuse natural-light withering conditions to complete the 48 h white-tea withering process. Because no duration-gradient experiment was conducted, this duration should not be interpreted as an optimized treatment duration.

Four treatments were designed as follows:

- CK (control): withering under indoor diffuse natural light for 48 h, followed by drying at 80°C for 2 h.
- R (red light): withering under LED red light (610–730 nm) for 3.5 h, followed by indoor diffuse natural-light withering for 44.5 h, and then drying at 80°C for 2 h.
- Y (yellow light): withering under LED yellow light (570–590 nm) for 3.5 h, followed by indoor diffuse natural-light withering for 44.5 h, and then drying at 80°C for 2 h.
- B (blue light): withering under LED blue light (450–460 nm) for 3.5 h, followed by indoor diffuse natural-light withering for 44.5 h, and then drying at 80°C for 2 h.

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After drying, the processed tea samples were sealed and stored at low temperature until further analysis.

2.2 | Sensory Evaluation

Following the Chinese National Standard GB/T 23776-2018, 3.0 g of tea was infused in 150 mL of boiling water (tea-to-water ratio 1:50) for 5 min. A panel of eight trained tea experts independently evaluated infusion color, aroma, and taste, assigning scores out of 100 for each attribute. Dry-leaf appearance and infused-leaf appearance were not included. The total score was calculated according to the proportionally converted weighting of the evaluated sensory attributes based on GB/T 23776-2018: aroma = 5/13, taste = 6/13, and infusion color = 2/13. Total score = aroma × 5/13 + taste × 6/13 + infusion color × 2/13, with higher scores representing superior sensory quality.

2.3 | Quantification of Taste Quality and Compounds

2.3.1 | E-Tongue Analysis

Taste profiles were analyzed using an SA-402B E-tongue (Insent, Kyoto, Japan) according to optimized protocols (S. Y. Tian et al. 2024). One gram of tea powder was brewed with 100 mL of boiling ultrapure water, filtered, cooled, and centrifuged (8000 rpm, 5 min) to obtain a clear supernatant.

Sensors for sourness (CAO), saltiness (CTO), umami (AAE), bitterness (COO), and astringency (AE1) were calibrated with standard reference and electrode solutions. Each sample was tested four times; the first reading was discarded and the remaining averaged. The taste potential was expressed as the difference between sample and reference signals.

2.3.2 | HPLC

Catechins, caffeine, and theanine were quantified using an LC-40B XR HPLC system (Shimadzu, Kyoto, Japan) in accordance with Chinese National Standards GB/T 8313-2018, GB/T 8312-2013, and GB/T 23193-2017. Tea polyphenols and catechins were extracted with 70% methanol at 70°C and analyzed using a C18 column at 278 nm. Caffeine was determined at 280 nm and theanine at 210 nm, both using external standard calibration.

2.3.3 | Ultraviolet-Visible (UV-Vis) Spectroscopy Assays

According to the Chinese National Standard GB/T 8313-2018, the content of tea polyphenols was determined. The hydroxyl groups of tea polyphenols are oxidized by Folin phenol reagent and quantified using a UV-Vis Spectrophotometer (Shimadzu, Kyoto, Japan) at a maximum absorption wavelength of 765 nm using gallic acid to prepare a working solution.

According to the Chinese National Standard GB/T 8314-2013, the content of free amino acids was determined by reacting phosphate buffer solution and indene ketone solution with amino acids in tea powder under boiling water conditions to form a purple complex. The absorbance is measured at 570 nm using the spectrophotometer, and free amino acids are quantified using theanine as a standard curve.

The anthrone colorimetric method was used to determine the soluble sugar content in tea leaves. Anthrone and concentrated sulfuric acid react with sugars in tea soup to form complexes. To prevent overheating, an ice bath was used, and the absorbance is measured at 620 nm using the spectrophotometer. Glucose was used to prepare a standard solution for quantification, and the results were expressed in glucose equivalents.

2.4 | Quantification of Aroma Quality and Compounds

2.4.1 | E-Nose Analysis

Aroma characteristics were evaluated using a PEN3 E-nose (Airsense Analytics, Schwerin, Germany), with optimized parameters (J. J. Tian et al. 2025). Tea infusion (1:50, w/v) was analyzed for 200 s, and response data from 195 to 200 s were used for quantitative comparison. Sensor responses were expressed as G/G0. Because the PEN3 MOS sensors have broad and overlapping response ranges, the E-nose data were interpreted as pattern-based indicators of overall aroma differences rather than as compound-specific measurements; compound-level interpretation was based primarily on the GC-IMS results. The nominal response preferences of the sensors were as follows: W1C, aromatic compounds; W5S, broad-range responses including nitrogen oxides; W3C, ammonia and aromatic compounds; W6S, hydrogen; W5C, alkanes, aromatic compounds and less-polar compounds; W1S, methane with a broad response range; W1W, sulfur-containing compounds; W2S, alcohols and partially aromatic compounds; W2W, aromatic and sulfur-containing organic compounds; and W3S, methane–aliphatic compounds (H. R. Huang et al. 2024; Xiang et al. 2024).

2.4.2 | GC-IMS

Volatile compounds were identified and quantified using a FlavourSpec GC-IMS system (G.A.S., Dortmund, Germany; Cao et al. 2024). One gram of tea powder and 1 mL of 2-methyl-3-heptanone (internal standard) were sealed in a 20 mL headspace vial. 2-Methyl-3-heptanone was used as the internal standard because of its stable GC-IMS response and suitability for semi-quantitative normalization of tea headspace volatiles, following previous tea volatile-analysis protocols (Zeng et al. 2025). Separation was performed on an MXT-5 column (15 m × 0.53 mm × 1 μm). Compounds were identified based on retention index (RI) and drift time (DT) relative to standards, and quantified by internal standard ratio. Volatile levels were expressed as internal-standard-normalized relative contents.

2.5 | Data Processing and Statistical Analysis

All experiments were performed in triplicate unless otherwise stated, and data are presented as mean ± SD. Statistical analyses

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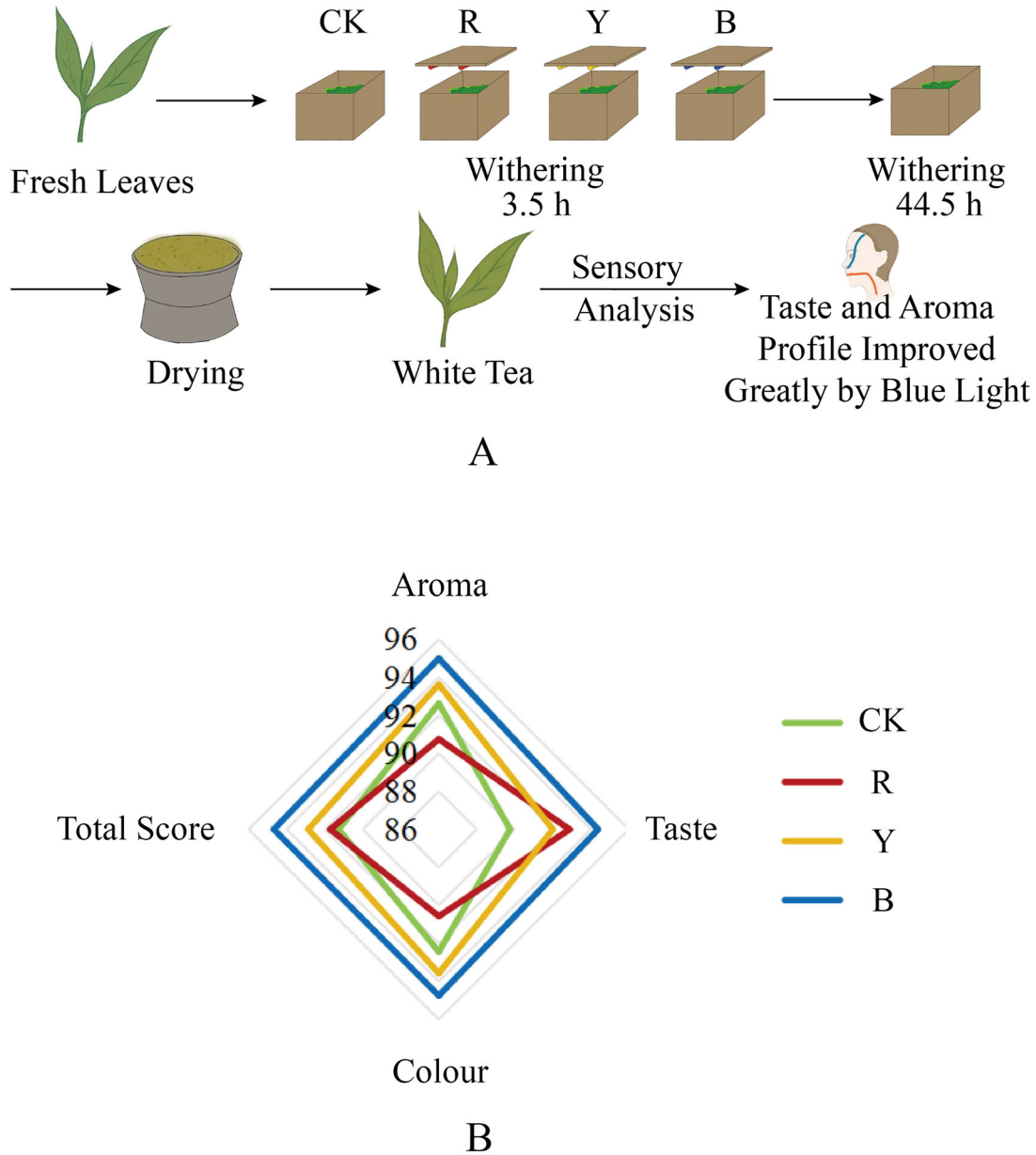


FIGURE 1 | Enhancing quality through different light quality pretreatments in autumn white tea. (A) Processing workflow and (B) sensory evaluation radar chart.

were conducted using SPSS 26.0 (IBM, Armonk, NY, USA). Differences among groups were evaluated by one-way analysis of variance (ANOVA), followed by Duncan's multiple range test for multiple comparisons. Differences were considered statistically significant at $p < 0.05$. Data visualization was performed using OriginPro 2021 for bar charts, TBtools for heatmaps, and Chiplot (https://www.chiplot.online/) for Mantel test plots.

3 | Results and Discussion

3.1 | Sensory Evaluation of White Tea Processed Under Different Light Qualities

To examine how light quality during withering affects the sensory characteristics of autumn white tea, four types of tea were prepared as illustrated in Figure 1A. Fresh leaves were withered under natural light (CK) or under red (R), yellow (Y), or blue (B) LED illumination for 3.5 h, followed by natural-light withering for another 44.5 h.

Sensory evaluation was conducted according to GB/T 23776-2018, using a panel of eight trained experts who assessed infusion color, aroma, and taste. The corresponding scores are summarized in Figure 1B and Table S1.

Among all treatments, the blue-light pretreatment group (B) displayed the most desirable sensory performance. The infusion exhibited a bright yellow color, a sweet and mellow taste, and a lasting caramel-like aroma. Its scores for aroma (95.00), taste (94.38), and infusion color (94.75) were the highest among all groups. The yellow-light group (Y) also improved sensory quality,

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though to a lesser degree, while the red-light treatment (R) resulted in slightly diminished aroma and color scores, 1.88 and 1.87 points lower than the control (CK), respectively.

Overall, the sensory evaluation results followed the trend B > Y > CK > R, indicating that blue-light pretreatment most effectively enhanced the overall flavor and visual quality of autumn white tea. The improvement likely relates to blue light's stimulatory effect on secondary metabolite biosynthesis, which contributes to both sweetness and aroma complexity.

3.2 | Taste Analysis of White Tea Under Different Early Light-Quality Pretreatments During Withering

3.2.1 | Quantitative Evaluation of Taste Intensity

Taste characteristics of autumn white tea processed under different light qualities were analyzed using an SA-402B E-tongue (Insent, Kyoto, Japan). Six key taste attributes were evaluated, including bitterness, astringency, umami, aftertaste-A, aftertaste-B, and richness (Figure 2). Saltiness and sourness responses were also recorded (Figure S1). These indices were not emphasized in the main text because previous white-tea sensory studies and tea taste reviews generally identify umami, sweetness/mellowness, bitterness, and astringency as the major quality-related taste descriptors, whereas saltiness and sourness are not typical core descriptors of white tea taste (Yu et al. 2025; Zheng et al. 2025). In addition, the sourness response in the present samples was far below the reported electronic-tongue tasteless point, indicating negligible sourness intensity; saltiness was retained as supplementary information but was not interpreted as a major quality-driving attribute (J. Yan et al. 2025).

Bitterness followed the order Y > R > B > CK (Figure 2A), whereas aftertaste-B showed the opposite trend (CK > R > B > Y, Figure 2B), indicating contrasting variation patterns between the two indices. Yellow-light pretreatment significantly increased bitterness while reducing aftertaste-B. Blue-light treatment also lowered aftertaste-B, compared with CK, which may be related to the more favorable taste performance of this group.

For astringency and aftertaste-A, the variation trends were Y > R > B > CK and Y > CK > R > B, respectively (Figure 2C,D). All light treatments increased astringency relative to CK. Yellow light significantly increased aftertaste-A, whereas red and blue light reduced it, suggesting that different wavelengths exerted distinct effects on residual taste perception.

Marked differences were also observed for umami and richness, both of which followed the order B > R > Y > CK (Figure 2E,F). The simultaneous increase in these two parameters in the blue-light group was consistent with its higher sensory taste score, suggesting an improvement in freshness and body of the tea infusion.

The relatively high sensory taste score of the Y group should be interpreted together with the E-tongue data. Although the Y group showed stronger instrumental bitterness and astringency responses than CK, its umami and richness responses were also higher than those of CK, while its aftertaste-B response was significantly lower. These results indicate that yellow light changed multiple taste attributes rather than simply deteriorating the taste quality.

Trained-panel taste scoring reflects an integrated perception of taste balance, aftertaste, and possible aroma–taste interactions during tea tasting, whereas the E-tongue provides separate instrumental response indices for individual taste attributes (Cho and Moazzem 2022; Hayashi et al. 2008; Moreira et al. 2024). Therefore, the lower bitter aftertaste, together with the higher umami/richness responses and honey/caramel-like retronasal aroma notes of the Y group, may have partly alleviated the negative contribution of increased bitterness and astringency to the overall sensory taste perception. In contrast, the B group showed the most balanced improvement across sensory scores, E-tongue taste attributes, and volatile composition, indicating that blue-light treatment more clearly improved the overall taste quality of autumn white tea. PLS-DA based on the electronic-tongue data showed clear separation among the four treatment groups (Figure 2G), indicating that taste profiles were effectively distinguished under different light-quality conditions. In the score plot, CK samples were all located in the fourth quadrant, whereas Y samples were all located in the first quadrant, showing a clear separation from CK. This distribution indicated that yellow-light withering induced the most pronounced shift in the instrumental taste profile, compared with the control, which was consistent with its higher bitterness and astringency responses and lower aftertaste-B response. B and R samples were mainly distributed between the second and third quadrants, indicating that blue- and red-light treatments also altered the taste profile but followed a different direction from yellow light. Notably, R samples were closer to the center of the score plot, suggesting that red light caused a relatively moderate taste-profile change, whereas B showed a more distinct shift associated with increased umami and richness and reduced aftertaste responses.

The model exhibited good explanatory and predictive ability ($R^2X = 0.995$, $R^2Y = 0.979$, $Q^2 = 0.918$), and the 200-permutation test confirmed that the model was reliable and not overfitted (Figure 2H). Among the taste variables, astringency and aftertaste-B showed VIP > 1 (Figure 2I), indicating that they were the main contributors to discrimination among the different light treatments. This result suggested that the separation among treatments was mainly driven by changes in astringent sensation and bitter aftertaste rather than by umami or richness alone.

When combined with the quantitative analysis of taste-active compounds (Figure 3), the PLS-DA and VIP results further indicated that light-quality withering modified the overall balance of taste-related metabolites. The stronger bitterness and astringency responses in the Y group were consistent with its relatively high contents of caffeine, epigallocatechin gallate (EGCG), epicatechin gallate (ECG), and C, which are important contributors to bitter and astringent taste in tea infusions. In contrast, the B group showed the highest levels of free amino acids and soluble sugars, which may partly explain its higher umami and richness responses. Although B also showed the highest EGC content, its lower GCG and C contents, together with the accumulation of amino acids and sugars, may have contributed to a more balanced taste profile by reducing the dominance of

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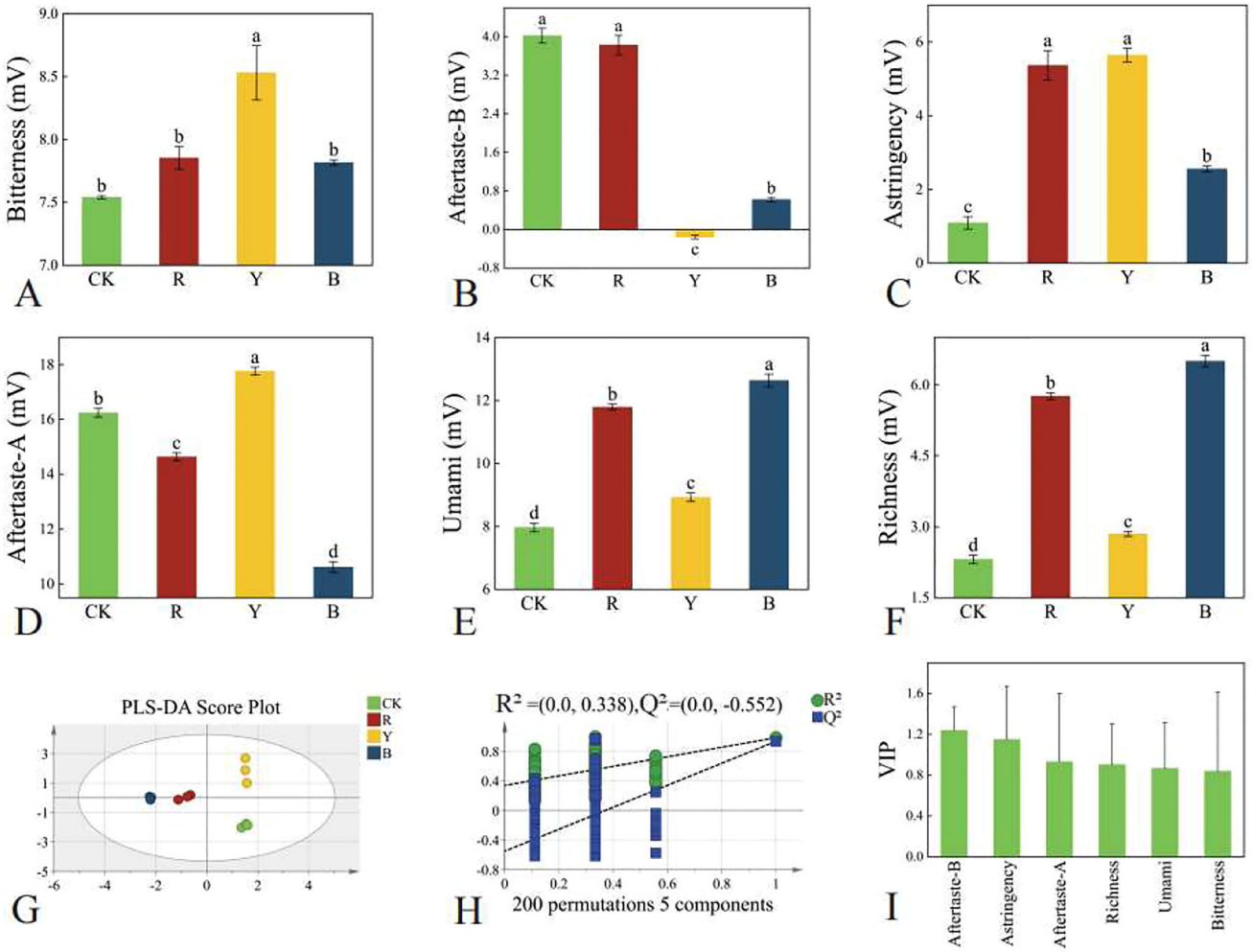


FIGURE 2 | Electronic tongue (E-tongue) analysis of different light quality pretreatments in autumn white tea. (A) Bitterness Intensity, (B) aftertaste-B Intensity, (C) astringency Intensity, (D) aftertaste-A Intensity, (E) umami intensity, (F) richness intensity, (G) partial least squares discriminant analysis (PLS-DA) score plot, (H) 200-permutation test for PLS-DA model validation, and (I) variable importance in projection (VIP) scores for sensors (VIP > 1). Data are presented as mean ± SD (biological replicates, $n = 3$; technical replicates: each sample was measured four times, and the average was used). One-way analysis of variance (ANOVA) followed by Duncan's multiple range test was performed. Different letters indicate statistically significant differences ($p < 0.05$).

bitter/astringent aftertaste. Therefore, the electronic-tongue PLS-DA separation reflected not only differences in individual taste indices but also changes in the balance among bitter, astringent, umami, and mellow taste components induced by different light qualities.

3.2.2 | Quantitative Analysis of Taste-Active Compounds

To clarify the major chemical basis underlying the differences in taste, representative key taste-active compound classes were selected to cover the major taste dimensions of white tea and quantified by HPLC and UV-Vis spectrophotometry (Qi et al. 2024). Specifically, caffeine, tea polyphenols, and individual catechins represent bitterness- and astringency-related compounds; total free amino acids and L-theanine indicate umami and freshness—L-theanine, a characteristic and abundant free amino acid in tea, is particularly associated with umami taste; and total soluble sugars reflect sweetness and mellowness (Juneja et al. 1999). Additional taste-active components, including catechin (C), tea polyphenols, catechin gallate (CG), and L-theanine, are shown in Figure S2.

The contents of free amino acids and soluble sugars increased significantly under light-quality treatments and followed the order B > R > Y > CK (Figure 3A,B), indicating that light treatments favored the accumulation of compounds related to umami and sweetness. The increase in free amino acids and soluble sugars may be associated with light-regulated carbon and nitrogen metabolism during withering. Tea leaves remain metabolically active after harvest, and light-quality signals can affect the degradation and transformation of proteins, starch, and other carbohydrate reserves (Ntezimana et al. 2024). Thus, blue-light pretreatment may promote nitrogen remobilization and protein degradation, contributing to free amino acid accumulation, while also enhancing starch or carbohydrate hydrolysis and sugar metabolism, resulting in higher soluble sugar levels. These changes are consistent with previous reports that light-assisted

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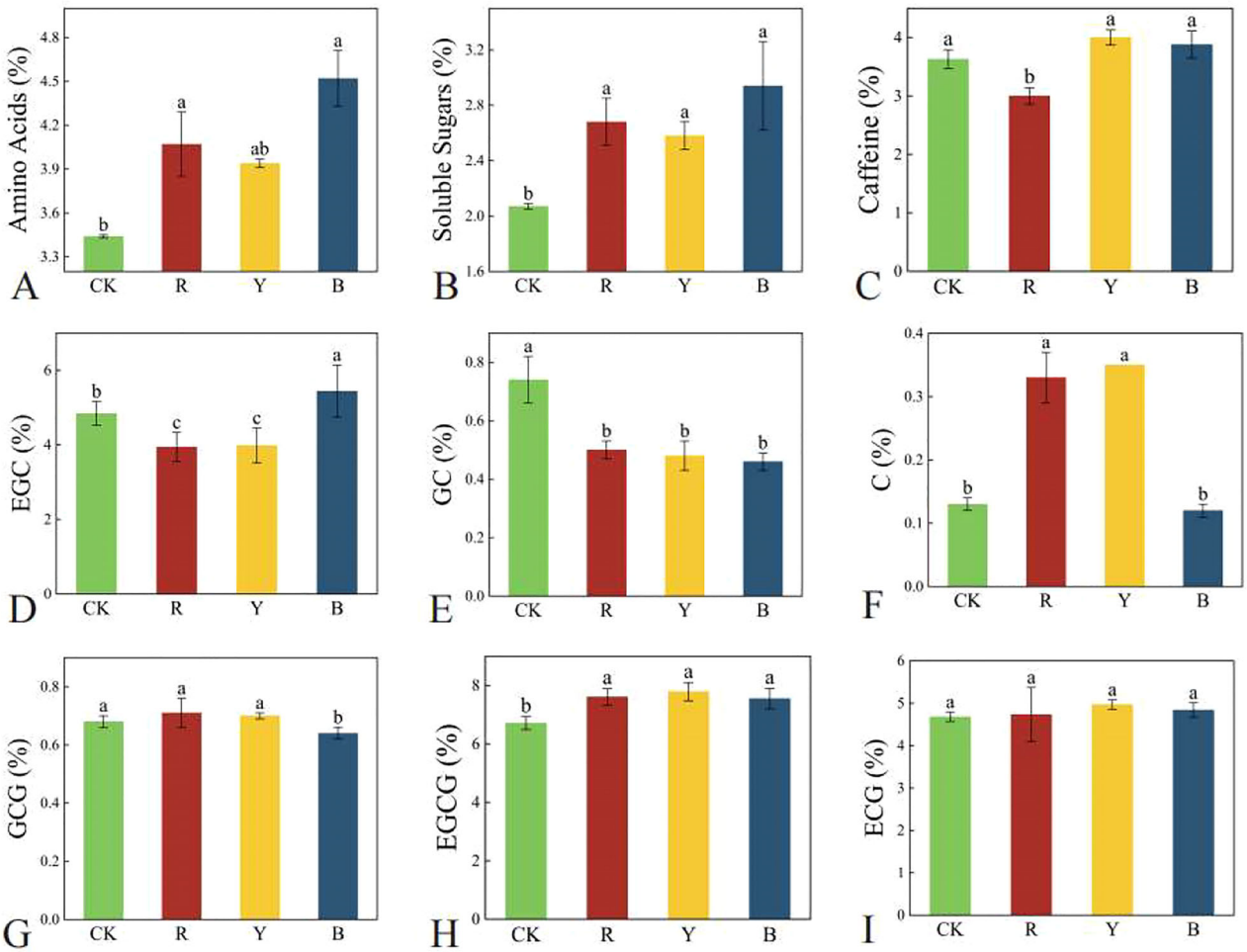


FIGURE 3 | Taste substance analysis of different light quality pretreatments in autumn white tea. (A) Amino acids content, (B) soluble sugars content, (C) caffeine content, (D) EGC content, (E) GC content, (F) C content, (G) GCG content, (H) EGCG content, and (I) ECG content. Data are presented as mean ± SD ($n = 3$). One-way ANOVA followed by Duncan's multiple range test was performed. Different letters above the bars indicate statistically significant differences ($p < 0.05$). For the Y group in Figure 3F, the original replicate values were 0.345825, 0.348100, and 0.350100, giving a non-zero SD; however, the SD becomes 0.00 when displayed to two decimal places, making the error bar visually indistinguishable.

withering and LED spectral regulation influence flavor-related metabolites in tea leaves, including amino acids and sugars (R. Huang et al. 2024; Q. M. Li et al. 2025).

In contrast, bitter- and astringency-related compounds showed different variation patterns. Compared with CK, the R group showed lower contents of caffeine, EGC, and GC but a higher content of C (Figure 3C–F). The Y group also showed reduced EGC and GC but elevated C. The B group exhibited the highest EGC content but the lowest GCG and C contents (Figure 3G), suggesting that blue light altered catechin composition during withering.

Overall, the chemical results were generally consistent with the E-tongue data. The relatively high contents of caffeine, EGCG, ECG, and C in the Y group were consistent with its stronger bitterness-related response, whereas the higher total free amino acid and soluble sugar levels in the B group may partly explain its higher umami and richness values (Figure 3H,I). These results suggest that the improved taste quality under blue-light pretreatment was associated with changes in major taste-active compound classes and catechin composition.

3.3 | Aroma Analysis of White Tea Under Different Early Light-Quality Pretreatments During Withering

3.3.1 | Quantitative Evaluation of Aroma Intensity

Among all sensors, W1C, W3C, and W5C showed the response pattern Y > B > R > CK (Figure 4A–C), indicating treatment-dependent changes in these broad aroma-response dimensions. This trend was consistent with the relatively high aroma scores of the Y and B groups in the sensory evaluation. However, because of the cross-sensitivity of MOS sensors, these responses should be regarded as indicators of changes in the overall volatile profile rather than evidence for increases in individual compound classes.

In contrast, sensors W1S, W2S, W2W, W3S, W5S, and W6S displayed the opposite trend CK > R > B > Y (Figure 4D–I). W1S and W3S are broadly associated with methane/short-chain alkanes and alkane-related volatiles, respectively. Their relatively high responses in CK and R suggest that these groups retained

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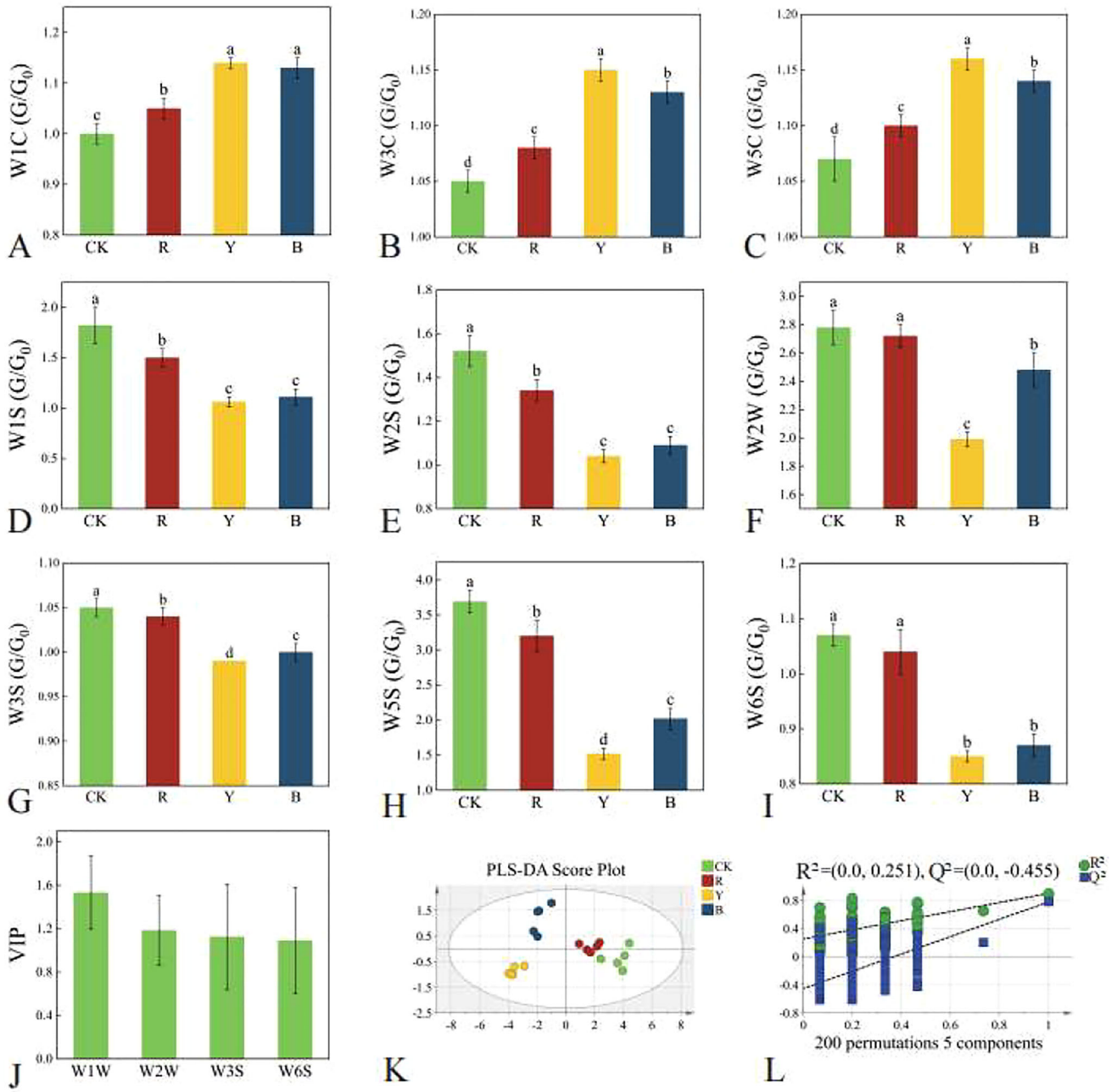


FIGURE 4 | E-nose analysis of different light quality pretreatments in autumn white tea. (A) W1C intensity, (B) W3C intensity, (C) W5C intensity, (D) W1S intensity, (E) W2S intensity, (F) W2W intensity, (G) W3S intensity, (H) W5S intensity, (I) W6S intensity, (J) VIP scores, (K) PLS-DA score, and (L) 200-permutation test. Data are presented as mean ± SD ($n = 5$). One-way ANOVA followed by Duncan's multiple range test was performed. Different letters indicate statistically significant differences ($p < 0.05$).

stronger hydrocarbon-related background signals. When considered together with sensory evaluation and GC-IMS results, this pattern may be associated with the persistence of green-like or less-mature aroma characteristics in these treatments. By comparison, the lower responses in B and Y indicate that light-quality withering, especially yellow-light treatment, reduced these broad sensor responses.

The responses of W2S, W2W, W5S, and W6S were also interpreted cautiously. W2S is broadly responsive to alcohols, aldehydes, and ketones, which may contribute to fruity, floral, green, or fatty notes depending on the specific compounds present. W2W is associated with aromatic compounds and organic sulfur compounds, which may be related to floral, roasted, or sulfur-like aroma notes. W5S and W6S are broad-response sensors associated with nitrogen oxides and hydrides/hydrogen, respectively, and therefore were used only as general indicators of changes in the volatile profile. Overall, these sensor responses indicated that light-quality withering altered the global aroma pattern of tea infusions, with yellow-light treatment producing the most distinct aroma-related response profile.

PLS-DA based on the electronic-nose responses showed clear separation among all treatment groups (Figure 4K), indicat-

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ing that the E-nose effectively discriminated aroma differences induced by light quality. In the score plot, CK samples were mainly located in the fourth quadrant, whereas R samples were distributed between the first and fourth quadrants, suggesting that red-light treatment caused a moderate shift in the aroma profile, compared with the control. By contrast, B samples were located in the second quadrant and Y samples in the third quadrant, clearly separated from CK and R along the first latent variable. This separation pattern suggested that blue- and yellow-light withering induced more pronounced changes in the overall volatile-response profile than red light, while the separation between B and Y along the second latent variable further indicated that blue and yellow light affected aroma formation through different volatile-response patterns.

The model showed acceptable performance ($R^2X = 0.995$, $R^2Y = 0.886$, $Q^2 = 0.783$), and the 200-permutation test confirmed the validity of the model without obvious overfitting (Figure 4L). VIP analysis identified W1W, W2W, W3S, and W6S as the major variables contributing to group discrimination (VIP > 1, Figure 4J). According to the broad response characteristics of the PEN3 sensor array, W1W is mainly associated with inorganic sulfur compounds (Figure S3), W2W with aromatic compounds and organic sulfur compounds, W3S with alkane-related volatiles, especially long-chain alkanes, and W6S with hydrides/hydrogen-related compounds. Therefore, the high VIP values of these sensors suggest that light-quality withering altered sulfur-related, aromatic/organic sulfur-related, hydrocarbon-related, and hydride-related volatile-response signals. These changes may reflect modifications in volatile metabolism during withering, including the transformation of green-like background volatiles and the formation of more characteristic aroma profiles under blue- and yellow-light treatments. Nevertheless, because MOS sensors have broad and overlapping selectivities, these VIP sensors should be interpreted as key contributors to the overall E-nose discrimination pattern rather than as selective markers of individual compounds.

Considering these results together with the GC-IMS analysis (Figure 5), the E-nose PLS-DA separation was consistent with compound-level volatile changes. In particular, yellow-light treatment was associated with increased floral- and fruity-related volatiles such as α-terpineol and linalool, whereas blue-light treatment enhanced sweet/roasted aroma-related compounds such as isoamyl acetate, ethyl acetate, furfural, and furaneol and reduced several green-note volatiles. Thus, the E-nose results provided pattern-level evidence for light-quality-induced aroma differentiation, while GC-IMS further explained these differences at the compound level.

3.3.2 | Qualitative and Quantitative Analysis of Aroma Compounds

To further characterize the volatile profiles of tea samples under different light treatments, volatile compounds were analyzed by GC-IMS, which allows sensitive separation and detection of trace aroma components. The two-dimensional topographic plots showed clear differences in volatile profiles among treatments (Figure 5A), with signal intensity roughly reflecting the relative abundance of compounds. Differential spectra relative to CK indicated that yellow- and blue-light treatments increased the abundance of most volatiles, whereas red light enhanced only a limited number of compounds.

A total of 103 volatile compounds were identified, including 29 alcohols, 19 aldehydes, 22 ketones, 12 esters, three furans, five pyrazines, eight terpenes, three hydrocarbons, and two others (Table S2). The yellow-light group (Y) was characterized by higher levels of volatiles such as α-terpineol, linalool, and 2-ethyl-1-hexanol, which are commonly associated with floral and fruity notes (Chen et al. 2020; J. L. Huang et al. 2024; H. Z. Lin et al. 2024). In contrast, the blue-light group (B) showed relatively high levels of isoamyl acetate, ethyl acetate, furfural, and furaneol, which are often associated with sweet, caramel-like, and roasted aroma characteristics. At the same time, blue light suppressed several green-note volatiles, including cis-3-hexenol, 1-hexen-3-one, and Z-ocimene (Jabłoński et al. 2024; L. Li et al. 2026; F. Yan et al. 2024; Q. Q. Li et al. 2024). These compositional changes were generally consistent with the sensory results, suggesting that blue light favored a sweeter and more complex aroma profile, whereas yellow light enhanced floral freshness.

PLS-DA based on the GC-IMS data clearly separated the four groups (Figure 5B), indicating that different light-quality treatments induced distinct volatile profiles. In the score plot, CK samples were all located in the first quadrant, whereas B samples were all located in the fourth quadrant, showing clear separation mainly along the second latent variable. This distribution suggested that blue-light treatment caused a marked shift in the volatile profile, compared with CK, which was consistent with the increased levels of sweet/roasted aroma-related compounds, such as isoamyl acetate, ethyl acetate, furfural, and furaneol, and the decreased levels of several green-note volatiles, including cis-3-hexenol, 1-hexen-3-one, and Z-ocimene. R samples were distributed near the Y-axis and were slightly closer to the second quadrant, indicating that red-light treatment caused a moderate alteration in volatile composition. In contrast, Y samples were located in the third quadrant but close to the X-axis, suggesting that yellow-light treatment produced a volatile pattern distinct from CK and B but less separated along the second latent variable. This separation was consistent with the higher levels of floral- and fruity-related volatiles, such as α-terpineol, linalool, and 2-ethyl-1-hexanol, in the Y group.

The model showed strong explanatory and predictive performance ($R^2X = 0.978$, $R^2Y = 0.989$, $Q^2 = 0.953$), and the permutation test further verified model stability (Figure 5C). GC-IMS identified 29 discriminatory volatile signals (VIP > 1), among which 12 showed VIP > 1.5, including furfural M, ethyl acetate M, furfural D, 1-octen-3-ol D, alpha-terpineol M, hexanal D, 1-butanol D, furaneol M, 3-methylbutanal, ethyl butanoate, 2-methylbutanal, and beta-pinene. (Figure 5D,E). Therefore, the PLS-DA and VIP results indicate that light-quality withering changed not only the total abundance of volatile compounds but also the balance among floral/fruity, sweet/roasted, and green-note volatiles. Specifically, blue light mainly promoted sweet and roasted aroma-related compounds while reducing green-note volatiles, whereas yellow light preferentially enhanced floral and fruity volatiles. Red light showed a relatively moderate effect on volatile remodeling.

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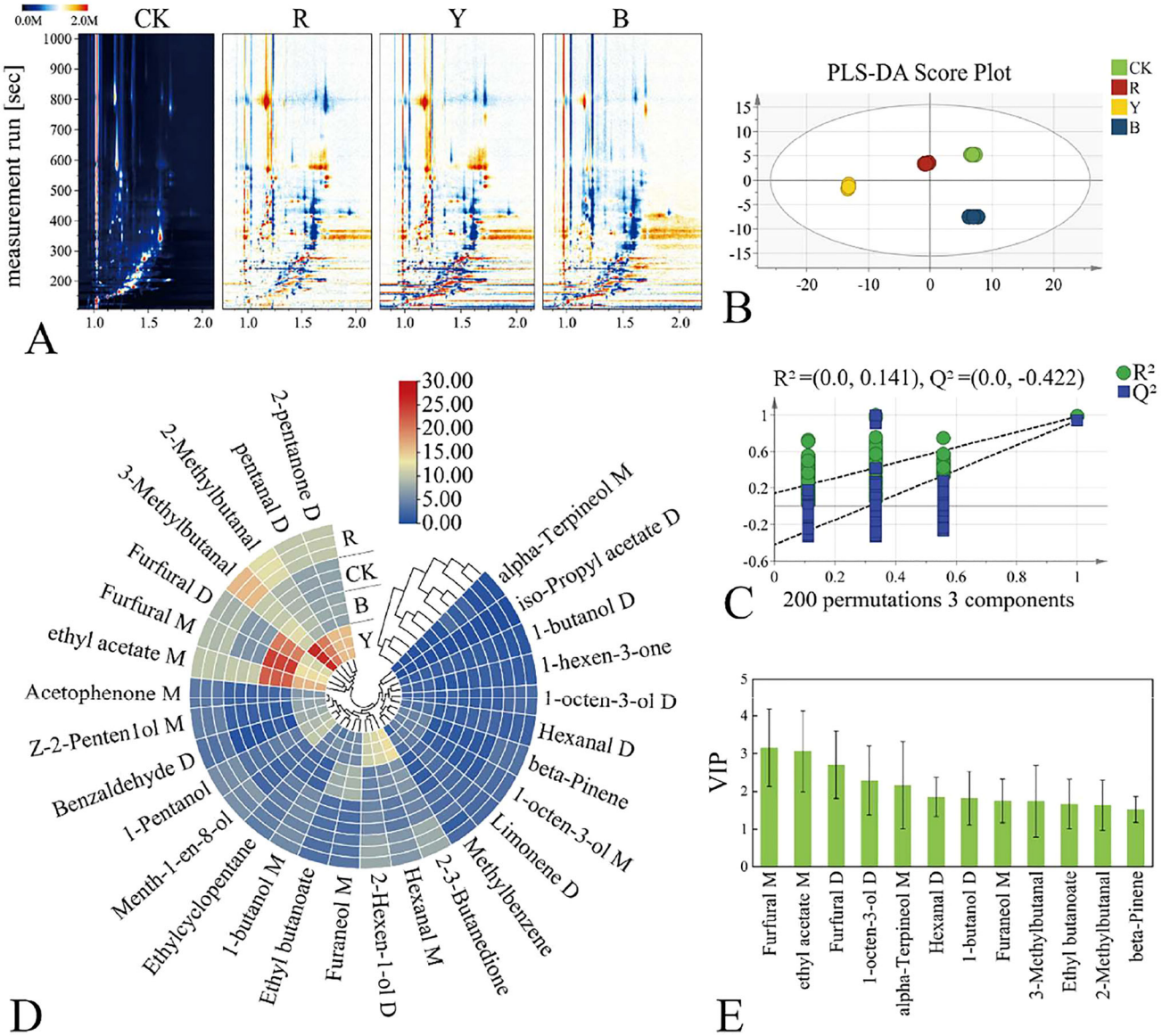


FIGURE 5 | GC-IMS analysis of different light quality pretreatments in autumn white tea. (A) Two-dimensional GC-IMS topographic comparison among treatment groups (CK, R, Y, B), (B) PLS-DA score plot based on GC-IMS features, (C) 200-permutation test for PLS-DA model, (D) Heatmap of 29 key aroma compounds, and (E) List of compounds with VIP > 1.5. *Note:* In GC-IMS compound labels, M, D, and T denote monomer, dimer, and trimer ion peaks, respectively; these ion forms were retained as separate signals when they showed distinct drift times.

Based on the identified volatile compounds and previously reported aroma formation mechanisms in tea, several possible formation pathways were inferred. Terpene alcohols, such as linalool and geraniol, may be released from glycosidically bound precursors through enzymatic hydrolysis, which has been reported to enhance the aroma of white tea infusion during β-glucosidase treatment (Ni et al. 2021). Branched-chain aldehydes, including 2-methylbutanal and 3-methylbutanal, are generally associated with amino acid-derived Strecker degradation during tea processing (R. J. Dong et al. 2023). In addition, C6/C8 alcohols and ketones, such as hexanol, 1-octen-3-ol, and related compounds, may originate from fatty-acid oxidation through the lipoxygenase–hydroperoxide lyase pathway, which has been recognized as an important route for lipid-derived volatile formation in green tea processing (Q. W. Wang et al. 2024). Furan compounds are likely related to sugar degradation and Maillard-related reactions during thermal processing, as reported in roasted Wuyi rock tea (Gao et al. 2025). It should be noted that these pathways were proposed based on the detected volatile profile and established tea aroma chemistry, and they were not directly confirmed by targeted pathway validation in the present study.

Taken together, early light-quality pretreatment during withering altered both the abundance and composition of volatile compounds in autumn white tea. Blue- and yellow-light treatments both improved aroma quality, although through different volatile patterns: Blue light mainly enhanced sweet and roasted aroma-related compounds while reducing green-note volatiles, whereas yellow light preferentially promoted floral and fruity volatiles.

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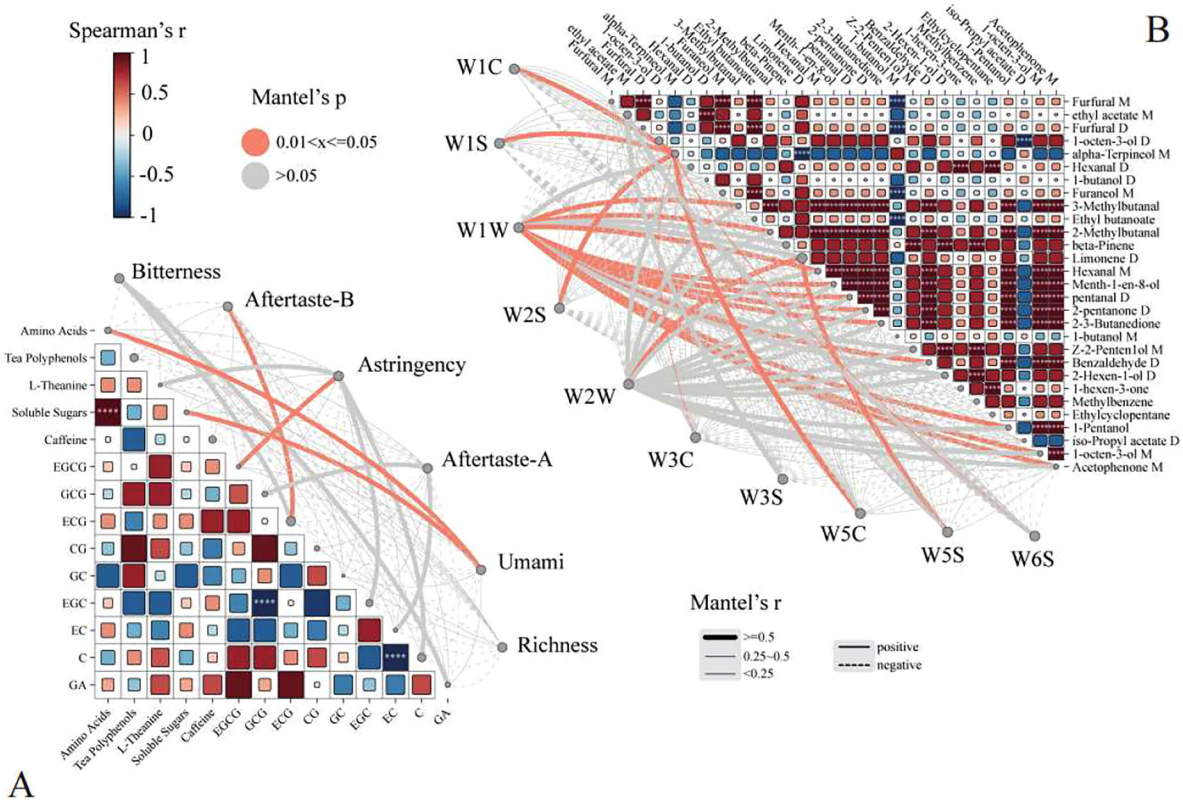


FIGURE 6 | Mantel test analysis of different light quality pretreatments in autumn white tea. (A) Mantel correlations between taste-active compound matrix and E-tongue responses (shown as correlation coefficients and significance), (B) Mantel correlations between GC-IMS volatile matrix (VIP > 1) and E-nose sensor signals. Interpretation: Mantel test detects overall matrix-level correlations; it does not imply direct causation.

3.4 | Integrative Associations Between Chemical Profiles and Sensor Readouts

To integrate chemical data with sensory-related instrumental responses, Mantel tests were performed between (i) the matrix of taste-active compounds and the electronic-tongue responses and (ii) the matrix of GC-IMS volatiles (VIP > 1) and the electronic-nose sensor signals (Figure 6).

Taste vs E-tongue. Significant Mantel correlations ($p < 0.05$) were observed between umami and free amino acids ($r = 0.556$), umami and soluble sugars ($r = 0.511$), astringency and EGCG ($r = 0.527$), and aftertaste-B and ECG ($r = 0.806$). These results were generally consistent with the known sensory functions of these compounds, namely, that free amino acids and soluble sugars contribute to freshness and body, whereas galloylated catechins are closely associated with astringency and lingering taste perception. The close association between ECG and aftertaste-B may be related to differences in polyphenol–protein interactions, although this requires further confirmation.

Volatiles vs E-noes. E-nose sensor W1W showed significant positive correlations with several alcohols, aldehydes, and ketones, including 3-methylbutanol, 2-methylbutanol, hexanal (M), pentanal (D), and 2,3-butanedione, suggesting that this sensor can serve as a useful indicator of changes in these volatile groups. W2S was positively correlated with α-terpineol ($r = 0.570$), and α-terpineol also showed positive associations with W1C, W1S, W2S, W3C, and W5C, supporting its contribution to floral-sweet aroma characteristics captured by multiple sensors. In addition, limonene (D) was correlated with W2W and W5S, which is consistent with the broad responsiveness of these sensors to terpenoid-related volatiles.

It should be noted that Mantel analysis detects overall matrix-level correlations rather than causative relationships. Moreover, because E-nose sensors show cross-sensitivity and volatiles may co-vary across treatments, individual sensors should be interpreted as broad indicators of aroma change rather than selective detectors of single compound classes.

Overall, the coupling between higher amino acid/sugar contents and stronger umami-richness responses, together with the association between volatile patterns and E-nose signals, supports the conclusion that blue- and yellow-light pretreatment improved flavor quality through coordinated changes in taste-active compounds and volatile profiles.

4 | Conclusion

This study demonstrated that light quality during withering significantly influenced the sensory quality and chemical characteristics of autumn white tea. Compared with natural-light withering (CK), different LED light treatments, especially blue and yellow light, altered both taste and aroma profiles, as revealed

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by integrated analyses combining sensory evaluation, E-tongue, E-nose, HPLC, UV-Vis spectrophotometry, and GC-IMS.

Among the tested treatments, blue-light pretreatment produced the best overall sensory quality, characterized by enhanced sweetness, umami, and richness, together with reduced bitterness and green-note-related characteristics. These changes were associated with increased levels of free amino acids and soluble sugars, decreased levels of certain catechins such as GCG and C, and reduced accumulation of green-note volatiles. Yellow light mainly improved aroma quality by promoting floral- and fruity-related volatiles, including linalool and α-terpineol, whereas red light showed limited positive effects.

GC-IMS analysis further showed that blue- and yellow-light treatments markedly reshaped volatile composition by increasing compounds related to sweet, roasted, floral, and fruity aroma characteristics, while suppressing several green-note compounds. Multivariate PLS-DA models showed good discrimination performance, and Mantel analysis further supported close associations between chemical changes and sensor-based taste/aroma responses.

Overall, this work establishes an integrated framework linking light quality, digital sensory responses, and chemical profiles in autumn white tea processing. From a practical perspective, blue-light pretreatment appears to be a promising strategy for improving the overall quality of autumn white tea. These findings also provide a basis for future optimization of LED light conditions, including wavelength, intensity, and treatment duration, to regulate taste and aroma formation in tea processing. Because the sensory panel size was limited and individual amino acid/sugar profiles, transcriptomic responses, enzyme activities, and continuous natural-light microclimate data were not measured, the proposed metabolic explanations should be regarded as plausible interpretations and require further targeted validation.

Author Contributions

Hujing Cao: methodology, investigation, data curation, formal analysis, visualization, Writing – original draft. **Jing Sun**: methodology, investigation, data curation, formal analysis, validation. **Yixin Song**: methodology, investigation, data curation, validation. **Haoyu Huang:** Investigation, validation. **Yan Zhu**: methodology, investigation, data curation, validation. **Xujun Zhu**: conceptualization, supervision, resources, funding acquisition. **Yue Wang**: supervision, resources, partial funding acquisition. **Jingjing Tian**: conceptualization, supervision, resources, funding acquisition, writing – review and editing. All authors meet the ICMJE authorship criteria. All authors participated in study design, data analysis, manuscript drafting, final approval, and agree to be accountable for all content of the manuscript.

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Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

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